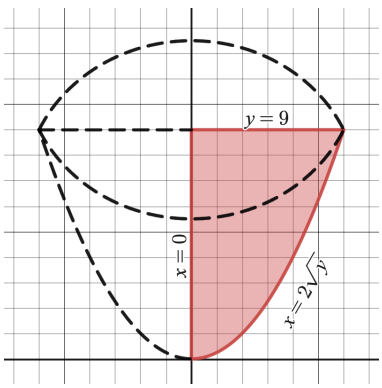
MATH 152 Discussion Problems: Feb. 5, 2024
Sections 02 & 04
Textbook: 6.2, 6.3

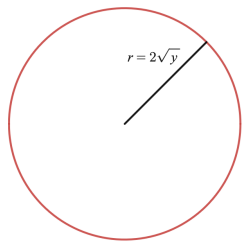
1. Use the disk method to find the volume of the solid obtained by revolving the region bounded by

$$x = 2\sqrt{y}\ ,\ x = 0\ ,\ \text{and}\ y = 9$$

around the y-axis.



cross section:



$$\begin{aligned} V &= \pi \int_a^b (\text{radius})^2 dy \\ &= \pi \int_0^9 (2\sqrt{y})^2 dy \\ &= \pi \int_0^9 4y dy \\ &= \pi\big(4 \cdot \frac{y^2}{2}\big)\Big|_0^9 \\ &= 2\pi y^2\Big|_0^9 \\ &= 2\pi(81 - 0) \\ &= 162\pi \end{aligned}$$

2. Do problem 1 using the cylindrical shells method: Find the volume of the solid obtained by revolving the region bounded by

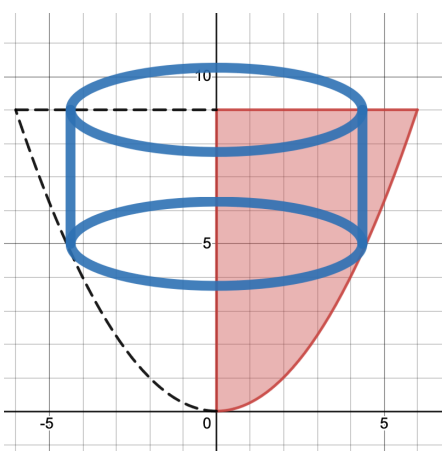
$$x = 2\sqrt{y}\ ,\ x = 0\ ,\ \text{and}\ y = 9$$

around the y-axis.

We will need $x = 2\sqrt{y}$ in terms of x: $y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4}$.

Curves intersect at $x = 0$ and when $9 = \frac{x^2}{4}$, so $x = \sqrt{9 \cdot 4} = 6$.

Thus, for each $x \in [0, 6]$, the shells have radius $= x$ and height $= 9 - \frac{x^2}{4}$.

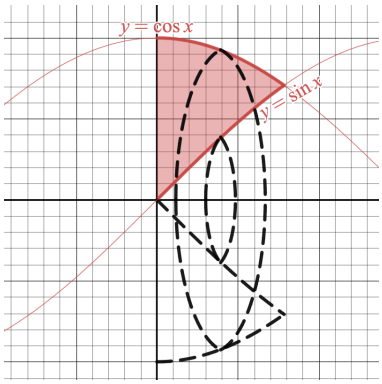


$$\begin{aligned} V &= 2\pi \int_a^b (\text{radius})(\text{height}) dx \\ &= 2\pi \int_0^6 x\big(9 - \frac{x^2}{4}\big) dx \\ &= 2\pi \int_0^6 \big(9x - \frac{x^3}{4}\big) dx \\ &= 2\pi\big(\frac{9x^2}{2} - \frac{x^4}{16}\big)\Big|_0^6 \\ &= \pi\big(9x^2 - \frac{x^4}{8}\big)\Big|_0^6 \\ &= \pi\big(9 \cdot 36 - \frac{6^4}{8}\big) \\ &= 162\pi \end{aligned}$$

3. Use the washer method to find the volume of the solid obtained by revolving the region bounded by

$$y = \sin x \text{ and } y = \cos x \ , \ 0 \le x \le \frac{\pi}{4}$$

around the x-axis.

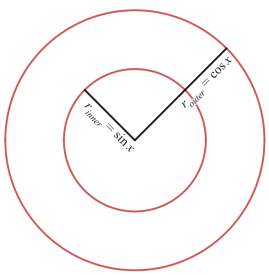


$$\begin{aligned} V &= \pi \int_a^b [(\text{outer radius})^2 - (\text{inner radius})^2] dy \\ &= \pi \int_0^{\frac{\pi}{4}} (\cos^2 x - \sin^2 x) dx \end{aligned}$$

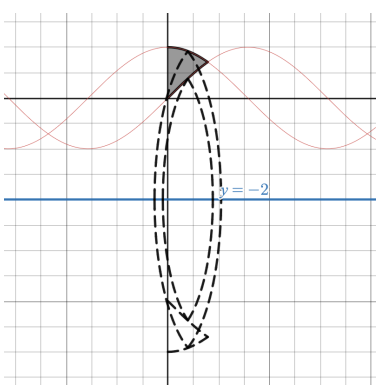
Use the double angle identity $\cos(2x) = \cos^2 x - \sin^2 x$.

$$\begin{aligned} &= \pi \int_0^{\frac{\pi}{4}} \cos(2x) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{4}} \cos(2x) d(2x) \\ &= \frac{\pi}{2} (\sin(2x))\Big|_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{2} \left(\sin\left(\frac{\pi}{2}\right) - \sin 0 \right) \\ &= \frac{\pi}{2}(1 - 0) \\ &= \frac{\pi}{2} \end{aligned}$$

cross section:



4. Repeat the previous problem but rotate the region around $y = -2$ instead of around the x-axis.

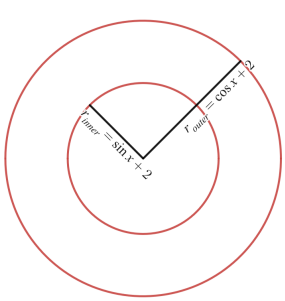


$$\begin{aligned} V &= \pi \int_a^b [(\text{outer radius})^2 - (\text{inner radius})^2] dy \\ &= \pi \int_0^{\frac{\pi}{4}} [(\cos x + 2)^2 - (\sin x + 2)^2] dx \\ &= \pi \int_0^{\frac{\pi}{4}} (\cos^2 x + 4\cos x + 4 - \sin^2 x - 4\sin x - 4) dx \end{aligned}$$

Use $\cos(2x) = \cos^2 x - \sin^2 x$.

$$\begin{aligned} &= \pi \int_0^{\frac{\pi}{4}} (\cos(2x) + 4\cos x - 4\sin x) dx \\ &= \pi \int_0^{\frac{\pi}{4}} \cos(2x) dx + \pi \int_0^{\frac{\pi}{4}} (4\cos x + 4 - 4\sin x) dx \\ &= \frac{\pi}{2} + \pi(4\sin x + 4\cos x)\Big|_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{2} + \pi\left(4\sin\left(\frac{\pi}{4}\right) - 4\sin 0 + 4\cos\left(\frac{\pi}{4}\right) - 4\cos 0\right) \\ &= \frac{\pi}{2} + \pi(2\sqrt{2} + 2\sqrt{2} - 4) \\ &= \left(4\sqrt{2} - \frac{7}{2}\right)\pi \end{aligned}$$

cross section:



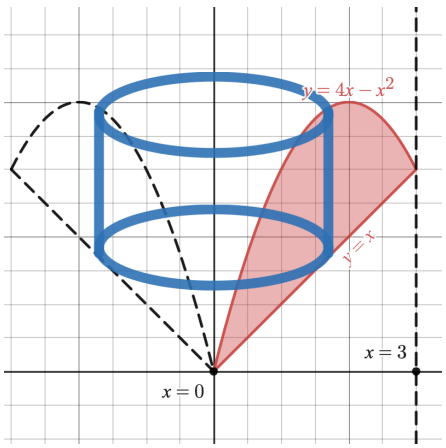
5. Use the cylindrical shell method to find the volume of the solid obtained by rotating the region bounded by

$$y = 4x - x^2 \text{ and } y = x$$

around the y-axis. (This is problem 14 in Ch. 6.3 of your textbook.)

First, find where these curves intersect:

$$\begin{aligned} 4x - x^2 &= x \\ 3x - x^2 &= 0 \\ x(3-x) &= 0 \\ x &= 0,\ 3 \end{aligned}$$



So, for each $x \in [0, 3]$, the shells have
radius $= x$
height $= 4x - x^2 - x = 3x - x^2$

$$\begin{aligned} V &= 2\pi \int_a^b (\text{radius})(\text{height})dx \\ &= 2\pi \int_0^3 x(3x - x^2)dx \\ &= 2\pi \int_0^3 (3x^2 - x^3)dx \\ &= 2\pi\left(3\frac{x^3}{3} - \frac{x^4}{4}\right)\Big|_0^3 \\ &= 2\pi\left(3^3 - \frac{3^4}{4}\right) \\ &= \frac{27\pi}{2} \end{aligned}$$